Rodrigo Gomez-Fell

Curriculum Vitae

Postdoctoral Fellow, Gateway Antarctica, University of Canterbury
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Education

- **2023** · PhD, Antarctic Studies, University of Canterbury, New Zealand
  Thesis: *Ice tongue stability in a changing climate*
- **2018** · MSc, Oceanography, Universidade Federal do Rio Grande, Brazil
- **2006** · Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Chile

Employment

- **2026–present** · Postdoctoral Fellow, Gateway Antarctica, University of Canterbury
- **2023–2026** · Geospatial Research Scientist, Waterways Centre, University of Canterbury
  Led the adaptation of the Riverscapes Framework and the Riverscapes Data Exchange to New Zealand as a national project
- **2023** · Research Assistant, University of Canterbury
- **2020–2023** · Teaching Assistant (part-time), University of Canterbury
- **2019–2023** · New Zealand Representative (part-time), DAP Airlines & Logistics
- **2018** · Researcher, CEQUA research centre, Chile
- **2014–2016** · Project Manager, CEQUA research centre, Chile

Awards and grants

Research grants and projects

- **2026** · Co-PI, Remote Sensing of Canterbury Hāpua Dynamics: delineation and analysis of hāpua change using a CNN model
- **2026** · AI in COSMO-SkyMed, Project ID 4362 Glaciers: tasking SAR data over the New Zealand Alps
- **2025–2027** · BIOMASS mission calibration and validation for Antarctic ice shelves, using pRES and airborne SAR (PI)
- **2025–2027** · Catalyst project: novel high-resolution imaging for glacial environments with polarimetric RADARSAT Constellation Mission data (AI)

Awards and scholarships

- **2023** · Best student presentation, SIRG Annual Meeting
- **2023** · Best student presentation, New Zealand Sea Ice Symposium
- **2023** · Scholarship to attend the IFT-R3S Summer School (Auckland, New Zealand)
- **2021–2022** · Antarctica New Zealand Sir Robin Irvine Doctoral Scholarship
- **2019–2022** · University of Canterbury Doctoral Scholarship
- **2017** · Scholarship to attend the POLENET Glacial Seismology Training School (Fort Collins, USA)
- **2016–2018** · Brazilian CNPq MSc Scholarship

Teaching and supervision

- **2020–2023** · Teaching Assistant, tutor and lab demonstrator (part-time), University of Canterbury: ANTA102, ANTA201, GEOG312, GEOG323 and GISC406
- **2026** · Delivered two SAR laboratory sessions at the 2nd IEEE GRSS NZ Radar Remote Sensing School, 15–17 April 2026
- Member of the supervisory team of a PhD student studying fast ice and ice shelf interactions in McMurdo Sound
- Helped supervise two MSc students at the Waterways Centre: mapping the Ashley braided river floodplain, and identifying hāpua with a CNN model
- Supervised three honours students working on New Zealand glaciers with SAR data

- **2023** · Mentor, Adopt a Scientist programme, Lincoln High School

Guest lectures in University of Canterbury courses

- **2021–present** · ANTA201: "Polar Remote Sensing" and "Ice tongues in the Western Ross Sea"
- **2022–present** · GISC406: "The Psychedelic world of Interferograms"
- **2023–present** · GEOG208: "Introduction to SAR" and "The value of lidar in New Zealand"
- **2026** · ENCN205 (Applied Data Analysis for Civil and Natural Systems): "Introduction to Remote Sensing"

Other guest lectures

- **2026** · Geomatics Department, Université de Sherbrooke, Québec, Canada (2 November 2026): "Radar and Optical Satellite Time Series of Rivers, Glaciers and Sea Ice in New Zealand and Antarctica"
- **2025** · AIUP USA undergraduate course (online): "Technology in Antarctica"
- **2025** · Aerospace Summit delegation at the University of Canterbury: "Scientific insight into New Zealand Riverscapes: From Satellites to Airborne Sensors"
- **2025** · Brazil delegation to the University of Canterbury: "From Airborne Sensors to Scientific Insight: The value of lidar in New Zealand"
- **2025** · Ministry for the Environment (online): "From Airborne Sensors to Scientific Insight: The value of lidar in New Zealand"
- **2025** · EOLab Auckland (online): "New Zealand Riverscapes: Uncovering River Change with Satellites and Lidar"
- **2025** · Icebreaker Xue Long II: "Ice tongue stability in a changing climate"
- **2024** · Wuhan University delegation: "Ice tongue stability in a changing climate"
- **2024** · Environment Canterbury (ECan): "Lidar Remote Sensing in New Zealand"

Invited talks and panels

- **2026** · Panellist, "Turning observation into action: How Earth Observation strengthens New Zealand's climate and economic resilience", InFocus sessions, New Zealand Aerospace Summit, Christchurch, 1 October 2026
- **2026** · Panellist, "Antarctica + Aerospace: From testbed to mission", InFocus sessions, New Zealand Aerospace Summit, Christchurch, 1 October 2026
- **2026** · Facilitated a discussion on investing in the next generation of Antarctic expertise, and presented a poster, at Christchurch's Antarctic Season Opening (21 September 2026), looking ahead to the Fifth International Polar Year (2032–33)
- **2026** · Invited talk as an early-career researcher, "Let's Talk Antarctica InSync" webinar, Antarctica InSync (online), 29 July 2026
- **2026** · Invited talk, Gateway Antarctica's Earth observation capabilities and research, DLR/NZ EU Horizon match-making event, 24 June 2026
- **2025** · Panellist, "Aerospace in Antarctica: The Ultimate Testbed", New Zealand Aerospace Summit, Christchurch, 7 October 2025

Conference presentations

Conference papers, abstracts and technical reports are listed at rgfell.github.io/publications.html and rgfell.github.io/reports.html.

Service

- **2027** · Session organiser (with Wolfgang Rack), Community Contributed Theme CCT.20, "Bridging Radioglaciology and Cryosphere Satellite Sensing", IGARSS 2027, Reykjavík
- **2026** · Chair, IEEE GRSS New Zealand Contiguous Chapter
- **2026** · Session Organiser, Theme Coordinator and Reviewer, IGARSS, Washington D.C.
- **2026** · Early-career reviewer, IPCC report
- **2025** · Secretary, IEEE GRSS New Zealand Contiguous Chapter
- **2025** · Session Organiser and Theme Coordinator, IGARSS, Brisbane
- **2025** · Coordinator, IGARSS Brisbane ambassadors for IGARSS 2028 Auckland
- **2025** · Organising Committee, Braided Rivers Forum
- **2024–present** · Member, IEEE GRSS IFT Working Group on Remote Sensing Instruments for UAVs
- **2024** · Organising Committee and Coordinator, New Zealand SAR Workshop (4 September 2024) and Hackathon
- **2024** · Reviewer, Antarctica New Zealand scholarships
- **2024** · Representative of the University of Canterbury, LatAm CAPES early-career delegation to Chile and Brazil

- **2023–present** · Member, IEEE Geoscience and Remote Sensing Society
- **2021–2023** · Member, International Glaciological Society Awards Committee
- **2019–2021** · Council Member, Association of Polar Early Career Scientists (APECS)
- Journal reviewer for *Nature*, *Journal of Glaciology* and *The Cryosphere*, among others